

INTERNSHIP PROJECT | PROJECT DOCUMENTATION

BuildTrack: Construction Project Management & Site Monitoring Platform



BuildTrack

— Construction Project Management & Site Monitoring —

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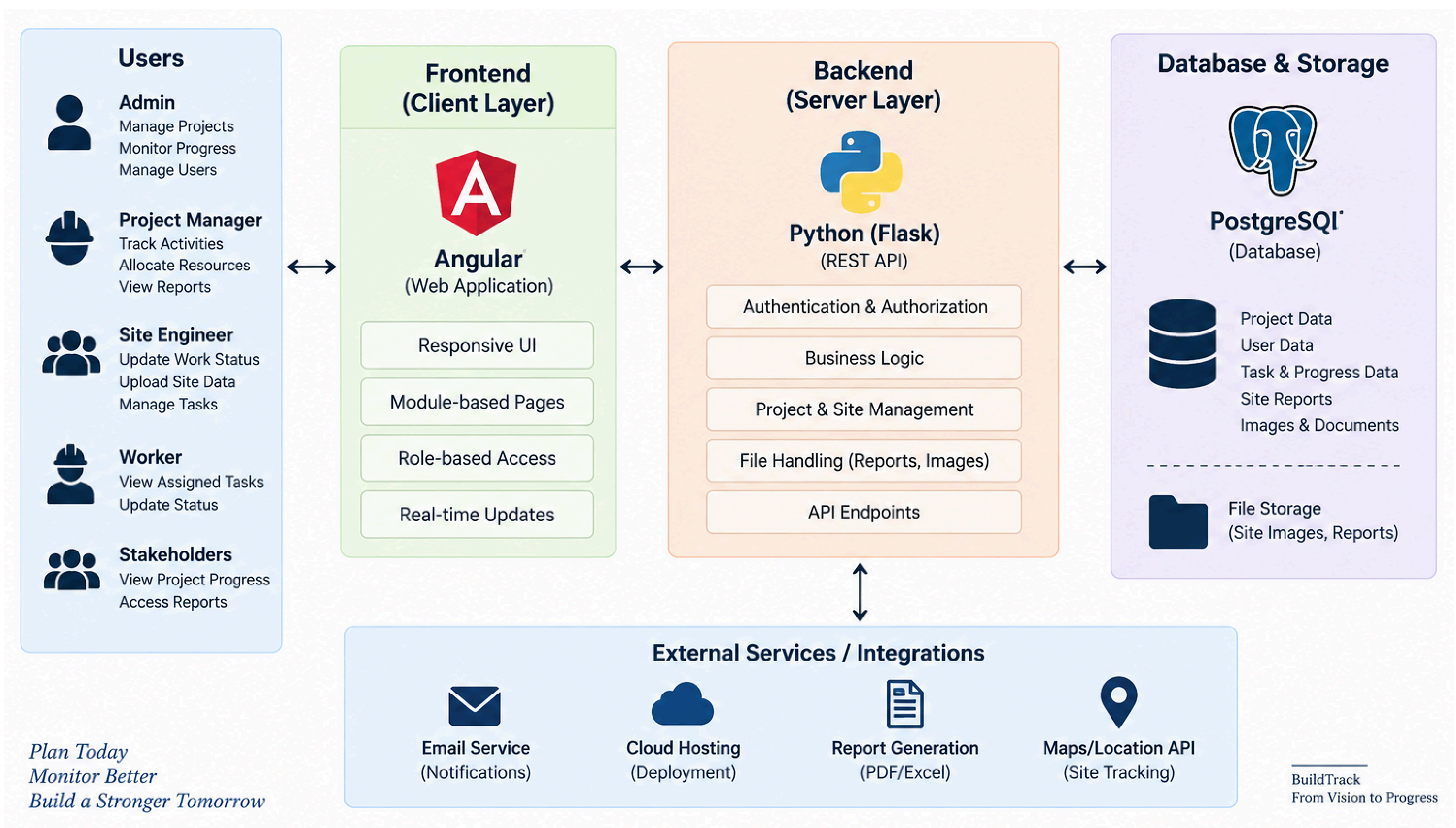
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1. PROJECT OVERVIEW

BuildTrack is a modern construction project management and site monitoring platform developed as part of the Infosys Springboard internship by Team 2. In typical civil infrastructure and commercial construction environments, site supervisors, project managers, contractors, and clients often face operational fragmentation due to disparate communication channels, lagging spreadsheets, and manual field paper logs. BuildTrack addresses these operational bottlenecks by bringing project activities, human capital, physical machinery, raw materials, procurement schedules, cost management, dynamic progress calculations, real-time alerts, executive dashboards, and exportable audit documentation into one unified digital system.

- Frontend: Angular, TypeScript, HTML and CSS
- Backend: Python with FastAPI
- Database: PostgreSQL / project development database
- ORM and database tools: SQLAlchemy and Alembic
- Security: JWT authentication, OAuth2 and role-based access control (RBAC)
- Architecture: service-oriented / clean-architecture approach with API, schema, security and business-logic layers



2. SYSTEM ARCHITECTURE

BuildTrack adheres to a modular, clean, layered architectural model separating presentation, business application logic, data persistence, and security controls.

Layer	Key Components & Responsibilities
Frontend (Angular)	TypeScript, Modular Components, HTML5, CSS3, Role-Based Route Guards, Interactive Dashboards, Dynamic Form Validation, Reactive State Management.
Backend (FastAPI)	Python 3, High-performance Async REST APIs, Pydantic Data Serialization, Dependency Injection for Auth, Modular Routing, Clean Separation of Service & Data Access Layers.
Security & Auth	OAuth2 Password Flow, Cryptographic JWT tokens, Role-Based Access Control (RBAC), Granular project-level access validation on all protected endpoints.
Database Layer	PostgreSQL relational engine managed via SQLAlchemy ORM; schema migration through Alembic; robust foreign key cascades across Projects, Users, Inventory, Workforce, and Budgets.

3. MILESTONE-WISE DEVELOPMENT SUMMARY

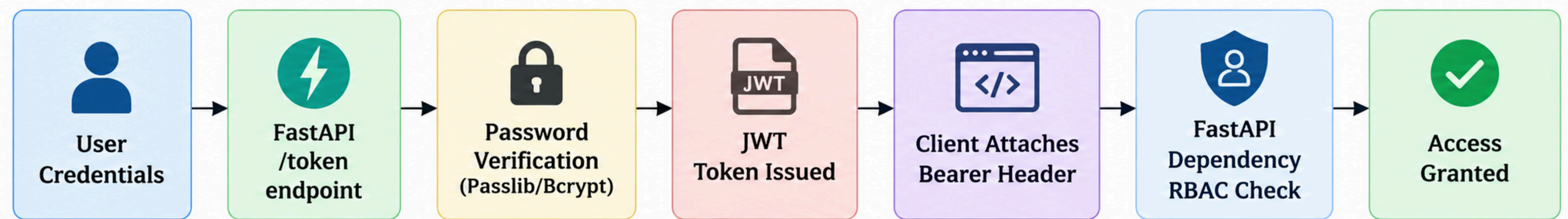
The platform was architected and realized across four consecutive, tightly synchronized sprint milestones:

Milestone	Modules Covered	Core Deliverables & Outcomes
Milestone 1 Foundation & Access	Module 1: Authentication & RBAC Module 2: Project Management	Established secure application foundation, JWT tokens, 6-role RBAC authorization hierarchy, project entity schemas, project manager associations, and baseline CRUD operations.
Milestone 2 Site Monitoring & Resources	Module 3: Site Progress Monitoring Module 4: Resource Management	Deployed daily progress report logs, milestone stage progress calculations, delay tracking taxonomy, equipment lifecycle tracking, allocation tables, and preventative maintenance logs.
Milestone 3 Materials, Workforce & Procurement	Module 5: Material & Inventory Module 6: Workforce Management Module 7: Procurement Management	End-to-end supply chain integration, stock movements, material requisitions, contractor/worker registries, shifts and attendance tracking, vendor catalogs, purchase orders, and invoice audits.
Milestone 4 Notifications, Analytics & Finance	Module 8: Notifications Module 9: Dashboards & Analytics Module 10: Reports & Docs Module 11: Budget & Cost Management	Complete business intelligence layer: automated event notifications, role-specific KPI dashboards, multi-module PDF/Excel reporting, cost categorization, and planned vs. actual budget variances.

4. MILESTONE 1 - MODULES 1 & 2

Module 1 - Authentication & Role-Based Access Control (RBAC)

Authentication verifies the user's identity, while authorization strictly controls what resources and actions an authenticated user can perform. BuildTrack enforces a 6-tier role model configured through secure JSON Web Tokens (JWT) containing cryptographically signed user payloads.



Supported Roles:

- **Admin:** Complete system-level configuration, user management, and audit visibility.
- **Project Manager:** End-to-end management of assigned projects, schedules, approvals, and budget allocation.
- **Site Engineer:** Field submission of daily site logs, machinery operational hours, and delay documentation.
- **Contractor:** Subcontractor oversight, work crew allocation, and trade-specific task reporting.
- **Worker:** Personal shift schedules, attendance recording, and assigned daily tasks.
- **Client:** Read-only executive progress inspection, financial summaries, and milestone audits.

Module 2 - Project Management

Project Management forms the master architectural anchor. All secondary entities (site logs, inventory allocations, work orders, expenses) relate to a distinct project ID. Key attributes include project code, title, designated Project Manager, start/target finish dates, active operational status, site location, and metadata configurations.

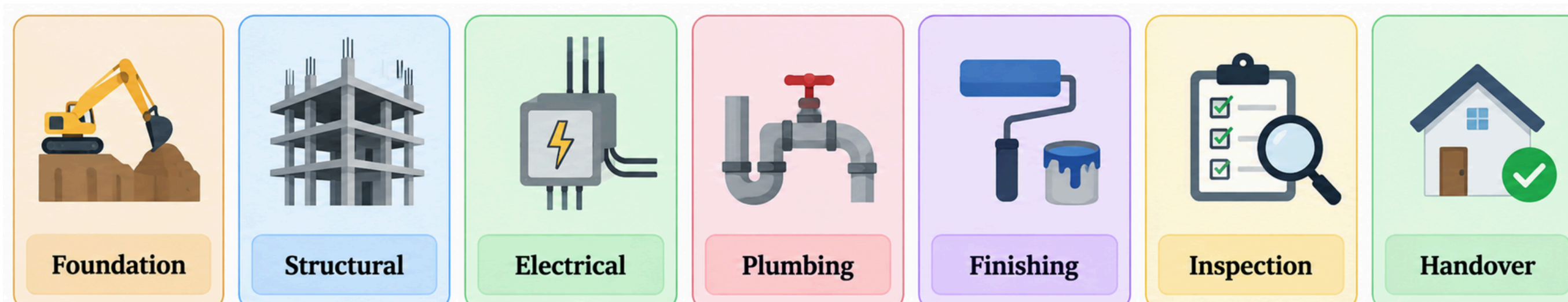
Milestone 1 Outcome: "Secure application foundation and project- management base established."

5. MILESTONE 2 - MODULES 3 & 4

Module 3 - Site Progress Monitoring

Module 3 captures ground reality on the construction site and converts raw field notes into quantifiable management telemetry.

- **Daily Progress Reports (DPR):** Captures date, work category, specific activity, completed percentage, contractor responsible, active headcounts, machinery deployed, weather conditions, safety audits, and material consumption.
- **Milestone Tracking:** Standardized construction phases:



- **Delay Tracking:** Root cause taxonomy tracking issues such as Heavy Rainfall, Labour Shortage, Material Delivery Delays, Machinery Breakdown, Design Modifications, Financial Blockers, and Government/Permit Approvals.
- **Automated Calculation:** Dynamic progress rollups weighting completed sub-tasks against overall project scope.

Module 4 - Resource Management

Manages heavy machinery and specialized tools, reducing idle downtime and preventing project delays through scheduled preventive maintenance.

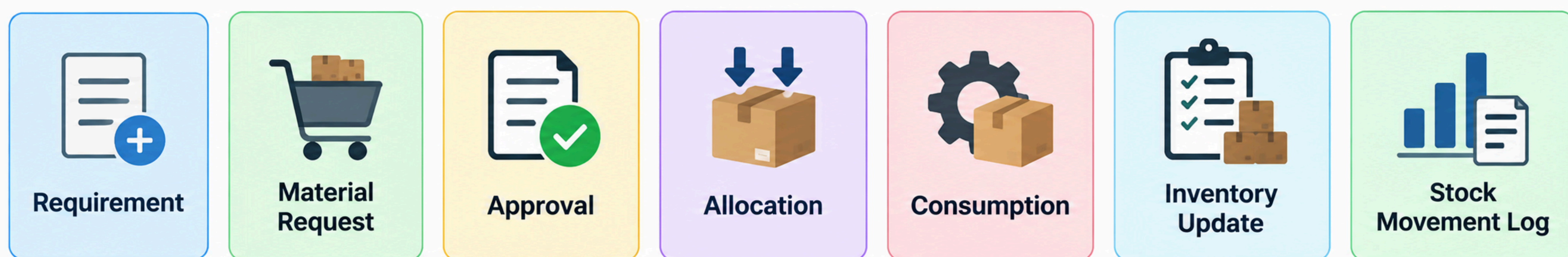
- **Equipment Catalog:** Categorization covering Excavators, Concrete Mixers, Tower/Mobile Cranes, Dump Trucks, Diesel Generators, and Safety Gear.
- **Equipment Status Life-Cycle:** Available • Allocated • Operating • Under Maintenance • Out of Service.
- **Utilization Tracking:** Logs running engine hours vs. standby/idle hours to identify underutilized capital assets.
- **Maintenance Hub:** Tracks service histories, scheduled inspection intervals, breakdown histories, and operating permits.

Milestone 2 Outcome: "Construction progress and resource monitoring became connected to project management."

6. MILESTONE 3 - MODULES 5, 6 & 7

Module 5 - Material & Inventory Management

Governs raw physical inventories across site storage yards, maintaining strict chain-of-custody across material requests, project engineer approvals, warehouse allocation, physical consumption, and automated stock deduction.



Maintains dynamic buckets for Available Stock, Allocated Stock, and Consumed Stock with threshold-based Low Stock alerts to prevent stockouts of critical items like cement, rebar, and aggregate.

Module 6 - Workforce Management

Coordinates site labour and contractor teams. Bridges individual workers to primary contractors, and contractors to project assignments.

- **Worker Registry & Categorization:** Engineers, Supervisors, Contractors, Skilled Workers, Unskilled Workers, and Specialist Consultants.
- **Operations Management:** Real-time daily attendance logging, shift scheduling (day/night shifts), site safety badge verification, and automated payroll rate calculations.

Module 7 - Procurement Management

Streamlines the requisition-to-procure pipeline. Features a centralized Vendor/Supplier Directory, formal Purchase Order generation, partial receipt workflows, warehouse delivery verification, and three-way invoice matching.

Milestone 3 Outcome: "Materials, workforce and procurement workflows were connected to project operations."

7. MILESTONE 4 - MODULES 8, 9, 10 & 11

Module 8 - Notification System

An event-driven notification hub keeping stakeholders updated on critical occurrences:

- **Alert Triggers:** Project updates, new work order assignments, purchase order approval notices, low-stock threshold triggers, delay notifications, and administrative alerts.
- **UI Implementation:** Top-bar notification bell with unread counters, sliding drawer panel, mark-as-read/mark-all-as-read controls, and role-filtered event streams.

Module 9 - Dashboard & Analytics

Provides role-tailored real-time analytical dashboards:

- **Project Manager View:** Overall percentage completion, cumulative cost burn-down, active contractor headcounts, machinery utilization rates, open delays, and upcoming milestone delivery dates.
- **Admin View:** Global system health, cross-project portfolio summaries, tenant user counts, and audit logs.

Module 10 - Reports & Documentation

Automates cross-module report compilation with parameterized filtering by project, date ranges, and functional categories. Generates standardized, audit-ready exports in PDF and Excel formats.

Module 11 - Budget & Cost Management

Delivers complete fiscal oversight tracking estimated budgets versus actual incurred project expenditures across 6 primary cost categories: Labor, Material, Equipment, Transportation, Maintenance, and Administrative overheads.

Milestone 4 Outcome: "BuildTrack gained its management, notification, analytics, reporting and financial decision-support layer."

8. OVERALL SYSTEM WORKFLOW

The entire operational pipeline of BuildTrack connects sequentially to ensure that high-level planning flows smoothly into site execution and back into analytical oversight:

Stage	Operational Mechanism
1. User Login	Credentials authenticated; role privileges decoded via JWT.
2. Project Initialization	Project entity instantiated with budget, milestones, PM assigned.
3. Resource & Workforce Setup	Contractors onboarded, workers assigned to shifts, equipment allocated.
4. Material & Procurement	Requisitions raised, POs approved, supplier deliveries stocked in inventory.
5. Site Execution & DPR	Daily field logs entered, machinery hours recorded, materials consumed.
6. Cost & Financial Rollup	Material consumption & labor shifts logged to expenses against planned budget.
7. Analytics, Alerts & Reporting	Event triggers dispatch notifications; dashboards update KPIs; exportable audits generated.

9. MODULE INTEGRATION

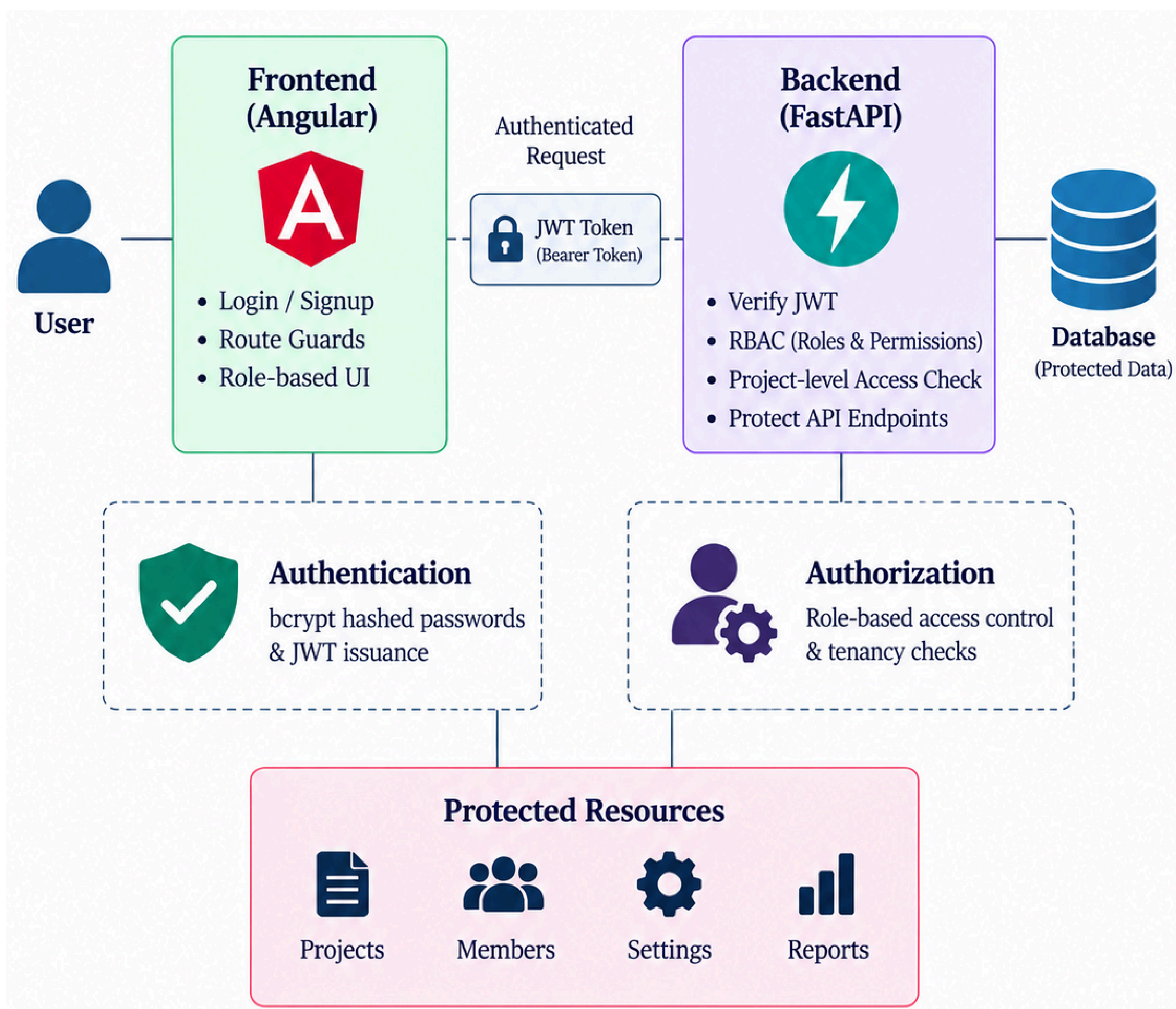
The system is designed around a central PROJECT entity which serves as the relational anchor. Data generated in one module automatically feeds into dependent modules:



10. SECURITY & ACCESS CONTROL

BuildTrack follows a defense-in-depth security approach, protecting the application at both the frontend and backend levels.

- **Authentication:** User identity is verified using securely hashed passwords with bcrypt and authenticated sessions are maintained using signed JWT tokens.
- **Role-Based Access Control (RBAC):** Access to features and APIs is controlled according to the user's assigned role and permissions.
- **Frontend Route Protection:** Angular route guards restrict unauthorized users from accessing protected pages and dynamically control available navigation options.
- **Backend Authorization:** FastAPI dependency-based security checks verify authentication and permissions before processing protected API requests.
- **Project-Level Access:** Users can access project data only when they have the appropriate project assignment or administrative permission.
- **Write Operation Verification:** Create, update, and delete operations perform backend authorization checks to prevent unauthorized data modification.
- **Secure Data Flow:** Communication between the Angular frontend and FastAPI backend uses authenticated API requests with bearer tokens, ensuring protected access to project data.



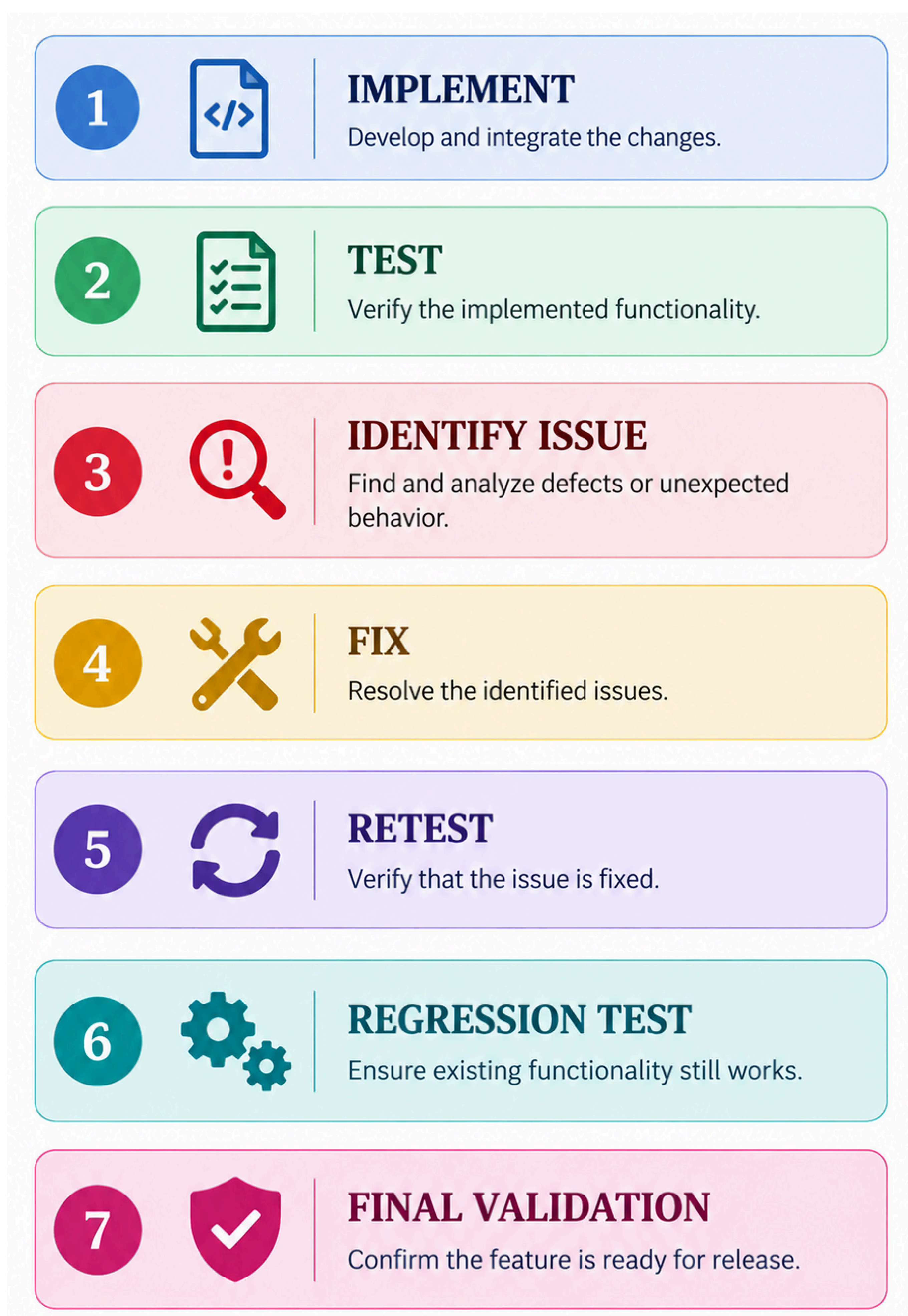
11. TECHNOLOGY STACK

Layer	Technology	Purpose
Frontend	Angular, TypeScript, HTML5, CSS3	Single-page application, reactive forms, role-tailored dashboards.
Backend API	Python, FastAPI	High-performance asynchronous RESTful APIs, business service layer.
Data Persistence	PostgreSQL, SQLAlchemy, Alembic	Enterprise relational database, Python ORM, schema migrations.
Security & Validation	JWT, OAuth2, Pydantic	Stateless authentication, schema request validation, and sanitization.
Architecture	Clean Architecture / Modular REST	Decoupled components, maintainability, scalable codebase.

12. TESTING & VALIDATION

A rigorous testing framework was instituted across development cycles to guarantee stability and prevent regressions:

- **API Testing:** Comprehensive endpoint validation covering HTTP status codes, payload structures, and error handlers.
- **Auth & RBAC Validation:** Validated that unauthorized users cannot execute protected actions via direct URL or manipulated API requests.
- **Form & CRUD Validation:** Enforced input validation, boundary condition testing, and foreign key constraint validations.
- **Integration Testing:** Verified cross-module transactions such as stock auto-deductions upon DPR material submissions.
- **UI & Responsive Testing:** Validated component layouts across desktop, tablet, and mobile displays, including dark mode support.



13. CURRENT PROJECT STATUS

All 11 modules have been implemented to the current project stage. The team is currently in the testing, integration, and final stabilization phase. A few interactive elements and edge-case workflows are being checked and refined during final testing to ensure enterprise-grade stability.

Validation Checklist	Status
All 11 Functional Modules Implemented	Completed
Cross-Module Inter-service Data Integration	Completed
FastAPI Backend API Validation & Schema Sync	Completed
PostgreSQL Relational Integrity & Cascades	Completed
Role-Based Access Control (RBAC) & Token Guards	Completed
Final UI Stabilization & Edge-Case Bug Fixing	In Final Polish

14. TEAM CONTRIBUTION

The project was planned, built, and delivered collaboratively by Team 2 with clear functional responsibilities:

Team Member	Designated Responsibilities
Shradha Sangita Dash	Frontend–Backend Integration
Netha Ushasri	Testing & Database
BV Sathvik Goparaju	Database & Dashboard UI
Gaurav Kumar	Backend & Business Logic
Abhishek Sardar	Responsive Design & Debugging
Jyoti Prakash Chindra	Research

14. FUTURE SCOPE

- **Mobile Application:** Develop Android/iOS apps for real-time site updates.
- **AI & Predictive Analytics:** Predict project delays, cost overruns, and resource requirements.
- **IoT Integration:** Connect sensors and equipment for real-time monitoring.
- **Advanced Dashboards:** Add more interactive analytics and performance insights.
- **Cloud Deployment:** Move the system to scalable cloud infrastructure.
- **Automated Notifications:** Provide smarter alerts for delays, low stock, payments, and deadlines.
- **Document Management:** Add centralized storage and automated project documentation.
- **Scalability:** Support larger projects, more users, and multiple construction sites.

15. CONCLUSION

BuildTrack successfully provides an integrated platform for managing construction projects by bringing project planning, site progress, resources, materials, workforce, procurement, notifications, dashboards, reports, and budget management into one system.

The project follows a secure, role-based architecture where each user can access features according to their responsibilities and project permissions. The integration between modules enables construction data to flow from daily site activities to progress tracking, inventory, workforce, procurement, financial monitoring, dashboards, and reports.

Overall, BuildTrack helps reduce manual work, scattered information, communication gaps, resource conflicts, material delays, and limited budget visibility. It provides project teams with a centralized and structured way to monitor construction activities and make better decisions.

With future enhancements such as AI-based predictions, IoT integration, mobile applications, cloud deployment, and advanced analytics, BuildTrack can be further developed into a scalable and intelligent construction management solution.



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